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LIS 440

Lecture: wk10: Lecture-Demo: More About Python and Pandas

11/04/25

Code

```
# Import the necessary packages
import pandas as pd # because 6 characters is to much
import os
from pathlib import Path # to set up paths easier

# Print version of pandas
print(pd.__version__)

# Set working directory (CWD) : root folder
root = Path(__file__).parent.parent.resolve()

# Show current working directory
print("Current Working Directory: ", Path.cwd())

# Set CWD: Data
data = root / 'data'
# Set CWD: Data > Raw
data_raw = data / 'raw'
# Set CWD: Data > Processed
data_processed = root / 'data' / 'processed'
# Set CWD: Scripts
scripts = root / 'scripts'

# [IGNORE] New dataframe Example
df = pd.DataFrame() # I don't like the arbitrarily long names Prof uses
print("Dimensions of this dataframe: ", df.shape)

# Read Student Data
student_data = data_processed / 'studentdatafile.csv'

# Create dataframe of student data
student_df = pd.read_csv(student_data)
# btw "\n" just puts it on newline
print("First 5 rows: \n", student_df.head()) # just show head
print("Last 5 rows: \n", student_df.tail()) # check if I am there

print("End of program.")
```

Shell

```
>>> %Run 11-4_thonny_demo.py
```

```
2.3.3
```

```
Current Working Directory: /Users/alexdelatorre/Library/CloudStorage/OneDrive-UW-Madison/Coursework/L I S  
440/Lectures/Week 10/Thursday/scripts
```

```
Dimensions of this dataframe: (0, 0)
```

```
First 5 rows:
```

	StudentNumber	LastName	FirstName	...	HomeState	Major	GPA
0	1	Pierce	Benjamin	...	Maine	Pre-Med	3.90
1	2	Van Pelt	Lucy	...	California	Psychology	2.80
2	3	Bueller	Ferris	...	Illinois	Undecided	4.00
3	4	Burns	Frank	...	Indiana	Pre-Med	1.00
4	5	Costanza	George	...	New York	Undecided	2.01

```
[5 rows x 8 columns]
```

```
Last 5 rows:
```

	StudentNumber	LastName	FirstName	...	HomeState	Major	GPA
81	82	McKessie	Taylor	...	Connecticut	Biology	4.00
82	83	Nielsen	Nielsen	...	New York	Music	3.80
83	84	Mendler	Bridgit	...	Massachusetts	Pre-Law	4.00
84	85	McGuire	Lizzie	...	California	Communications	3.85
85	86	De La Torre	Alejandro	...	California	Economics	3.34

```
[5 rows x 8 columns]
```

```
End of program.
```

```
>>>
```